MUHAMMAD AMMAR

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OBJECTIVE

As a passionate Software Engineer, I strive to make systems work smoothly and efficiently. Skilled in multiple programming languages and frameworks, I continuously seek opportunities to optimize performance and improve user experience. I am eager to contribute to innovative projects and make a meaningful impact through my technical abilities and problem-solving skills.

EDUCATION & CERTIFICATIONS

BS(Computer Science)
Comsat University Islamabad , Wah Campus

ICS
PUBJAB GROUP OF COLLEGES , SARGODHA

MATRIC
DAR-E-ARQAM MODEL SCHOOL , SARGODHA

KEY COMPETENCIES

- **Programming:**
  - Python, C++, Java, PHP, Dart, SQL
- **AI & Machine Learning:**
  - Retrieval-Augmented Generation (RAG), Semantic Embeddings, NLP, Plagiarism Detection Algorithms
- **Frameworks:**
  - Flutter, Laravel, Django, MERN Stack
- **Databases:**
  - MySQL, Firebase Firestore, Vector Databases
- **Web:**
  - HTML, CSS, Bootstrap, RESTful APIs, API Authentication (Sanctum)
- **Cybersecurity:**
  - CIA Triad, OSI Model, Risk Assessment, Vulnerability Management
- **Networking:**
  - IP Addressing, Subnetting, Routing, Switching (CCNA concepts)
- **Dev Tools:**
  - Git, Postman, VS Code, Android Studio, Asynchronous Processing (Background Jobs)
- **Cloud & Server Management:**
  - Basics of AWS & Windows Server (MCSE concepts)
- **Soft Skills:**
  - Problem Solving, Team Collaboration, Agile/Scrum Participation
- **Monitoring & Observability:**
  - Prometheus, Grafana (System & Application Metrics)
- **DevOps & Automation:**
  - Jenkins (CI/CD Pipelines), SonarQube (Code Quality & Security Gates)

INTERNSHIP EXPERIENCE

Flutter Developer Intern — Saturn Software House **Apr 2025 – June 2025**

- Collaborated on the development of cross-platform mobile apps using Flutter and Dart.
- Assisted in UI design, API integration, and bug fixes.
- Participated in daily standups and contributed to Agile sprint planning.
- Gained hands-on experience with Firebase and Git version control.

WORK EXPERIENCE

Projects **Sep 2022 - Present**

Weather App (Flutter & Dart)

- Built a weather application using Flutter and Dart.
- Integrated real-time weather API to fetch temperature, humidity, and weather conditions.
- Designed an interactive and responsive UI with smooth animations.

Esports Website (HTML, CSS, JavaScript, Django)

- Created a dynamic esports website using Django as the backend.
- Designed a responsive UI with HTML, CSS, and JavaScript.
- Implemented features such as tournament registration, player rankings, and live updates.

Quiz Game (Java)

- Developed a quiz game using Java with an interactive graphical user interface (GUI).
- Implemented question bank management, timer-based questions, and score tracking.
- Focused on performance optimization and engaging user interaction.

Snake Game (Python – Pygame)

- Developed a classic snake game using Python and the Pygame library.
- Implemented game mechanics such as collision detection, increasing difficulty, and score tracking.
- Designed a smooth and interactive user experience.

Car Rental App – (Flutter & Dart)

Developed a full-featured car rental application with separate interfaces for both renters and users.

- **Renter Side:**
  - Renters can add, update, or remove car listings including name, model, price, and optional images.
  - If no custom image is provided, the app automatically displays a default image based on car brand.
  - Renters receive real-time booking notifications from users.
  - Booking requests can be accepted or rejected by the renter, triggering automatic status updates.
- **User Side:**
  - Users can browse all available cars listed by different renters.
  - Car availability and booking status are updated dynamically.
  - Users can send booking requests and view their booking history.
- **Shared Features:**
  - Implemented tracker management: trackers move from "available" to "active" upon booking confirmation.
  - Built a responsive UI with bottom navigation linking Dashboard, Trackers, and Profile pages.
  - Focused on smooth state management, real-time updates using Firebase, and an intuitive user experience.

Smart Life Organizer (Flutter)

- Developed a productivity-focused mobile app using Flutter and Dart.
- Integrated features such as task management, note-taking, and contact organization.
- Added Firebase for cloud storage, real-time data syncing, and authentication.
- Designed a clean, user-friendly UI with interactive elements and responsive design.
- Ensured offline capabilities and performance optimization for a seamless experience.

Car Rental Website (HTML, CSS, JavaScript, Laravel)

- Created a car rental web platform using Laravel for the backend.
- Designed a dynamic frontend using HTML, CSS, and JavaScript with responsive layout.
- Implemented booking management, car listings, and real-time availability tracking.
- Added authentication, form validation, and database integration for secure data handling.
- Focused on a clean, modern UI with user-centric navigation and smooth interaction.

AI-Powered Learning Management System (Flutter, Laravel, Python)

- Engineered a full-stack enterprise educational platform featuring tailored portals for students and teachers.
- Developed a responsive Student Portal with real-time dynamic timetables, prerequisite-validated course registration, and automated grade tracking.
- Designed an administrative Teacher Portal for streamlined course roster management, exam scheduling, and secure grading synchronization.
- Integrated an AI Chat assistant using Retrieval-Augmented Generation (RAG) to provide intelligent, context-aware answers from uploaded course materials.
- Built a state-of-the-art Plagiarism and Collusion Detection engine using semantic embeddings to identify complex paraphrasing and ensure academic integrity.
- Implemented role-based authentication via Laravel Sanctum and utilized backend background jobs to process heavy machine learning tasks seamlessly.

Automated 3-Tier Application Deployment & Monitoring Pipeline

- Architecture Deployment: Designed and configured a scalable 3-tier application architecture to ensure strict separation of concerns among the presentation, application, and database layers.
- CI/CD Automation: Engineered a continuous integration pipeline using Jenkins to automate software builds and deployment workflows.
- Code Quality Assurance: Integrated SonarQube into the pipeline to enforce code quality gates, automatically detecting bugs, code smells, and security vulnerabilities prior to deployment.
- Observability & Monitoring: Implemented a robust monitoring stack using Prometheus and Grafana to collect and visualize real-time infrastructure metrics for both the Jenkins build machine and the application itself.

EXTRACURRICULAR ACTIVITIES

Courses

- MERN for Beginners
- Flutter Development Bootcamp with Dart (YouTube/Practical Labs)
- Complete Python Developer in 2022: Zero to Mastery (YouTube/Practical Labs)
- 100 Days of Code: The Complete Python Pro Bootcamp (YouTube/Practical Labs)
- HTML, CSS, and JavaScript for Web Developers (YouTube/Practical Labs)
- C++ Tutorial for Complete Beginners (YouTube/Practical Labs)
- CCNA – Cisco Certified Network Associate (YouTube/Practical Labs)
- MCSE – Microsoft Certified Solutions Expert (YouTube Learning)
- Cybersecurity Fundamentals (YouTube + Self Study)
- Cyber Risk Management (YouTube + Practice Scenarios)
- OSI Model – Computer Networking Basics (YouTube + Labs)

Hackathon

Problem Breakdown
Hitec, Islamabad

REFERENCES

References will be provided on demand